

JOSÉ ANDRÉS MORA

Phone: 416 294 0288 • Email: jmora.a@proton.me • GitHub: github.com/Jose-Andres-MoraLinkedIn: linkedin.com/in/joseandresmora • Portfolio: jose-andres-mora.github.io

SUMMARY OF SKILLS AND QUALIFICATIONS

Operating Systems | Windows • Linux • macOS • Bash • Unix command line **Licenses** | Valid driver's licence
Programming | C++ • Java • Python • TypeScript/JavaScript • HTML • JSON • ARM Cortex-M Assembly
Hardware and Embedded Systems | STM32 Nucleo-F334R8 • Arduino Uno • Raspberry Pi • Nexys A7 FPGA board
Engineering Tools | Vivado • ModelSim • STM32CubeMX • LabVIEW • NI DAQmx • LTspice • Oscilloscope • Multimeter
Development Tools | Visual Studio • VS Code • Keil µVision • IntelliJ IDEA • Eclipse • Git • GitHub • Vercel • Supabase
Languages | English | Spoken & Written – fluent • Spanish | Spoken & Written – fluent • French | Spoken – beginner

EDUCATION

Bachelor of Engineering – Computer Engineering Co-op **2024 - 2029 (Expected)**
Concordia University, Montreal, QC

- Member of the Institute for Co-operative Education
- Dean's List • Next-Generation Scholar Award – paid undergraduate research award (2026)
- GPA 3.90/4.30
- Relevant Courses: Data Structures + Algorithms • Digital Systems • Circuit Analysis • Object-Oriented Programming

Master of Fine Arts – Studio Arts **2018 - 2020**
University of Guelph, Guelph, ON | Dean's List • Ontario Graduate Studies – Research Scholarship

Bachelor of Fine Arts – Interdisciplinary Arts **2008 - 2013**
Nova Scotia College of Arts and Design | Dean's List

RESEARCH & WORK EXPERIENCE

Incoming Research Assistant — Next-Generation Scholar Program **May 2026 – Aug 2026**
Professor Paula Lago, Concordia University, Montreal, QC

- Selected to contribute to faculty-supervised research on semantic digital twin health systems.
- Research maps wearable sensor data to structured medical ontology concepts for health-risk interpretation.
- Responsibilities include reviewing cardiovascular and physical-activity ontologies, designing semantic mappings, and modeling uncertainty in wearable-data interpretation.
- Project will use Neo4j graph schemas and reasoning queries to connect synthetic wearable-data inputs to ontology-based health insights.

Technical Skills: Neo4j • Knowledge graphs • Medical ontologies • Wearable sensor data • Semantic modeling

Undergraduate Teaching Assistant — Digital Systems Design II Lab (COEN 313) **Jan 2026 – May 2026**
Lab Coordinator Tadeusz Obuchowicz, Concordia University, Montreal, QC

- Supervised labs involving VHDL, RTL simulation, logic synthesis, and FPGA implementation.
- Guided students through ModelSim simulations, test vectors, Vivado synthesis/implementation, XDC constraints, and Nexys A7 FPGA demonstrations.
- Assisted with debugging VHDL designs, including combinational logic, clocked circuits, shift registers, register logic, and simulation/synthesis mismatches.
- Evaluated lab demonstrations and helped students connect HDL simulation results to synthesized FPGA hardware.

Technical Skills: VHDL • RTL design • ModelSim • Vivado • Nexys A7 FPGA prototyping • XDC constraints • Digital logic

PROJECTS

Cryostat Temperature Data Logger

Sept 2025 - present

Research/Volunteer Project

Supervised by Professor Valter Zazubovits, Department of Physics, Concordia University

- Developed a LabVIEW-based data logging system to read, timestamp, and save temperature-related voltage data.
- Implemented CSV-based workflows for calibration tables, measurement logging, and averaging samples.
- Investigated signal behavior, input configuration, scaling, and verification strategies for reliable data collection.

Technical Skills: LabVIEW • NI DAQmx • Data logging • Sensor data • Voltage measurement

Flashcard Web Application

Apr 2026 - present

Personal Software Project

Developed a web-based flashcard application to organize course material, and support exam preparation.

- Built and deployed the app using TypeScript/JavaScript, GitHub, Vercel, and Supabase for backend data storage.
- Debugged deployment, routing, and repository configuration issues during the hosting process.

Technical Skills: TypeScript/JavaScript • Supabase • Vercel • GitHub • Web deployment • Database-backed applications

Battery Discharge Data Logger

Mar 2026 – Apr 2026

Academic Engineering Project

- Built a Nucleo-based voltage logging system to measure AAA battery discharge under a constant 33 Ω resistive load.
- Wrote Python scripts to log CSV data, convert timestamps to elapsed hours, plot discharge curves, and estimate delivered capacity in mAh through numerical integration.
- Improved experimental reliability by revising the test setup after observing unstable readings during faster discharge conditions.

Technical Skills: Nucleo microcontroller • Python • Data logging • Voltage measurement • Experimental testingMore details available at www.jose-andres-mora.github.io

PROFESSIONAL ASSOCIATIONS

Member: Electrical Team – Spacecraft Division

Oct 2024 – Feb 2025

Space Concordia, Concordia University, Montreal, QC

- Soldered and troubleshooted surface-mount components on a CubeSat PCB, supporting circuit verification.

VOLUNTEER WORK

Member of the Board of Directors

Nov 2022 – Aug 2023

VUCAVU, Toronto, ON

- Provided organizational guidance for nonprofit streaming platform aimed at promoting Canadian artists.

Mentor

Sept 2022 – Dec 2022

Creating Opportunities for Racial Equity Program, University of Guelph, Guelph

- Mentored a student pursuing a career in the visual arts by providing guidance on institutional navigation.

INTERESTS

Reading

Short-format Sci-fi: "Stories of Your Life" – Ted Chiang, "Frankenstein" – Mary Shelley

Hobbies

Running, drawing, learning cooking techniques, baking.

Passions

Visual arts, community-driven art festivals